Smart Employee Onboarding and Identity Service

1. Project Overview

- AWS-native digital onboarding system that automates the new-hire journey from employee creation to Day 1 readiness.
- Creates the canonical employee identity record that other HRMS services can reference.
- Automates Cognito identity provisioning, document collection and validation, workflow orchestration, HR notifications, progress tracking and web access.
- Uses serverless AWS services to minimize operational overhead and scale with onboarding volume.

2. Objectives

- Create a canonical employee profile with an automatically generated UUID.
- Provision a Cognito user identity for each new hire.
- Orchestrate Document Collection, IT Provisioning, Policy Sign-off and Manager Intro using AWS Step Functions.
- Collect and validate required onboarding documents in encrypted Amazon S3.
- Track employee, document and stage completion status in Amazon DynamoDB.
- Expose onboarding progress and workflow actions through Amazon API Gateway and AWS Lambda.
- Notify HR when all required documents are verified.
- Provide employee and HR web interfaces through Amazon S3 and Amazon CloudFront.

3. Solution Architecture

- The application follows an AWS-native serverless architecture with CloudFront and S3 for the frontend, Cognito for identity, API Gateway and Lambda for APIs, DynamoDB for state, Step Functions for workflow orchestration, S3 for documents, SNS/SES for notifications, CloudWatch for logs and CloudFormation for infrastructure as code.

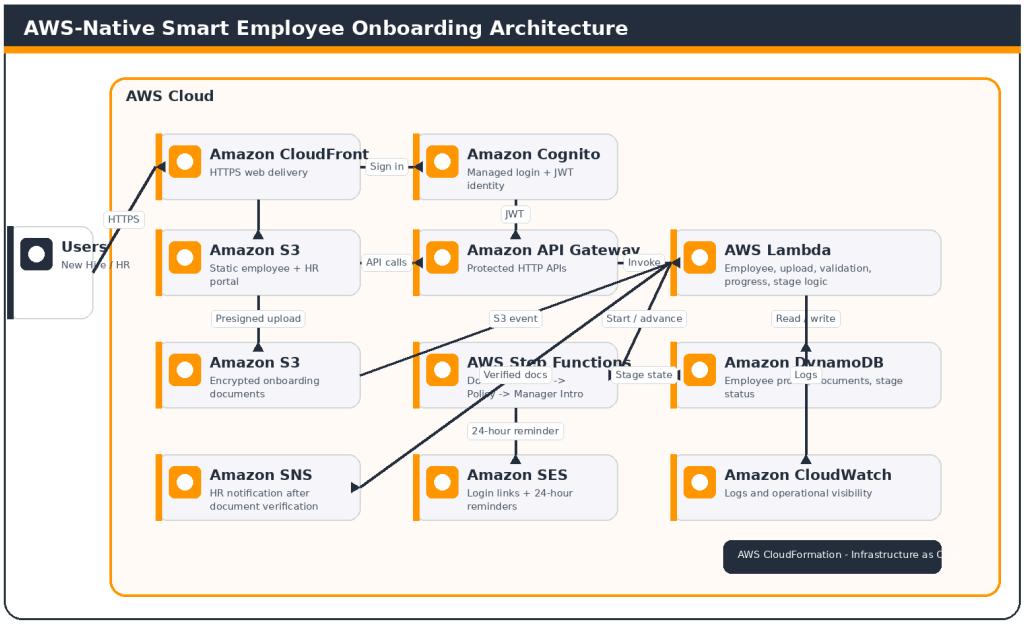


Figure 1. solution architecture

4. AWS Services Used

AWS CloudFormation: Infrastructure as code for core AWS resources.

AWS Lambda: Employee creation, upload URL generation, document validation, progress APIs, workflow helpers, reminders and completion logic.

Amazon DynamoDB: Single-table storage for employee profile, documents and onboarding stage state.

Amazon Cognito: New-hire identity, managed login and JWT issuance.

AWS Step Functions: Orchestrates the onboarding checklist and reminder loops.

Amazon S3: Encrypted document storage and static frontend hosting.

Amazon API Gateway: HTTP API endpoints and JWT-protected routes.

Amazon SNS: HR notification when all required documents are verified.

Amazon SES: Target delivery channel for login links and 24-hour reminders.

Amazon CloudFront: HTTPS delivery for the web frontend.

Amazon CloudWatch: Logs and operational observability.

5. Employee Record and Identity

- On employee form submission, the Create Employee Lambda generates a UUID employee_id.
- The Lambda stores the structured employee profile in DynamoDB.
- Structured attributes include name, email, department, role, manager, joining date, employment type and overall onboarding status.

- The workflow provisions a Cognito user and starts the Step Functions onboarding execution.
- Cognito Managed Login and API Gateway JWT authorization are working.
- SES-based login-link delivery remains a final integration item; Cognito account creation and login are already operational.

6. Onboarding Workflow Engine

- AWS Step Functions Standard models the required sequence: Document Collection -> IT Provisioning -> Policy Sign-off -> Manager Intro.
- Each stage is backed by Lambda logic and a DynamoDB stage record.
- Incomplete stages enter a wait-and-reminder loop.
- The demonstration workflow uses a 60-second wait so reminder behavior can be shown during a short demo.
- The production workflow definition uses 86400 seconds (24 hours), matching the assignment requirement.
- Reminder delivery currently uses SNS during testing; SES is the intended production delivery channel.

7. Document Collection and Validation

- Required documents are ID proof, degree certificate and signed offer letter.
- The frontend obtains S3 presigned URLs for direct document upload.
- The document bucket uses server-side encryption and is not publicly accessible.
- An S3 ObjectCreated event triggers the document-validation Lambda.
- The Lambda validates file type and file size and stores verification status in DynamoDB.
- Missing or incomplete documents prevent the DOCUMENTS stage from completing.
- When all required documents are verified, the DOCUMENTS stage is marked COMPLETE and HR receives an SNS notification.

8. Progress API and Frontend

- The New Hire Portal and HR Dashboard are static web applications hosted in Amazon S3 and delivered through Amazon CloudFront over HTTPS.
- The employee portal displays onboarding progress and stage status.
- The HR dashboard displays the onboarding pipeline and can complete supported IT Provisioning, Policy Sign-off and Manager Intro stages.
- Protected API routes use an API Gateway JWT authorizer connected to the Cognito user pool.
- An unauthenticated request to the progress endpoint returns HTTP 401 Unauthorized, confirming authentication enforcement.

9. DynamoDB ER Diagram

The logical data model contains one employee with multiple onboarding-stage records and multiple document records.

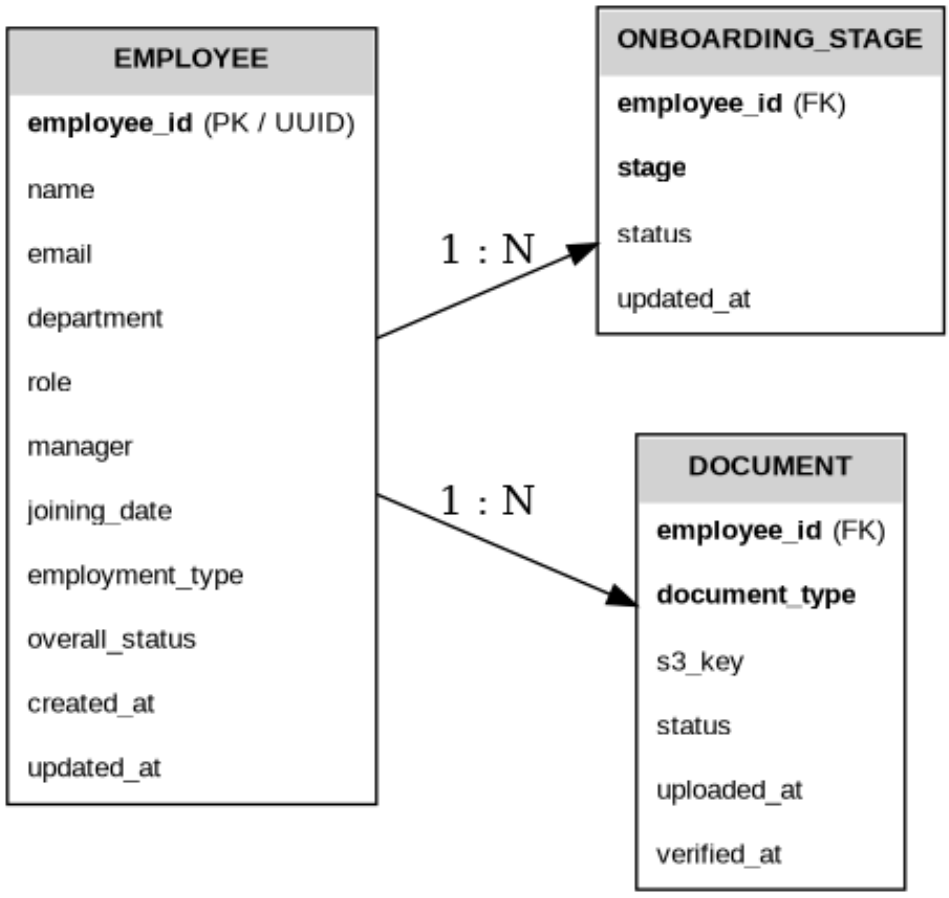


Figure 2. Logical employee, onboarding stage and document relationships

10. DynamoDB Single-Table Design

PK = EMPLOYEE#<employee_id> | SK = PROFILE
PK = EMPLOYEE#<employee_id> | SK = STAGE#DOCUMENTS
PK = EMPLOYEE#<employee_id> | SK = STAGE#IT_PROVISIONING
PK = EMPLOYEE#<employee_id> | SK = STAGE#POLICY_SIGNOFF
PK = EMPLOYEE#<employee_id> | SK = STAGE#MANAGER_INTRO
PK = EMPLOYEE#<employee_id> | SK = DOCUMENT#ID_PROOF
PK = EMPLOYEE#<employee_id> | SK = DOCUMENT#DEGREE_CERTIFICATE
PK = EMPLOYEE#<employee_id> | SK = DOCUMENT#OFFER_LETTER

11. Main API Endpoints

- POST /employees - Creates an employee profile and starts onboarding.
- POST /upload-url - Generates a presigned S3 document-upload URL.
- GET /employees/{employee_id}/progress - Returns employee details, documents, stage status and progress percentage.
- POST /employees/{employee_id}/stages/{stage}/complete - Marks a supported onboarding stage as complete.

12. Security

- Amazon Cognito provides managed authentication for onboarding users.
- Amazon API Gateway JWT authorization protects progress and stage-action routes.
- Amazon S3 document storage uses server-side encryption and public access blocking.
- Amazon DynamoDB uses server-side encryption.
- Amazon CloudFront provides HTTPS delivery for the frontend.
- AWS IAM execution roles apply service-specific permissions for Lambda and Step Functions.

13. Assignment Requirement Status

- UUID employee record in DynamoDB - Complete. Employee profile plus stage and document records are working.
- Cognito user account - Complete. Managed login tested successfully.
- SES login link - Partial. Cognito account/login works; SES login-link integration remains.
- 4-stage Step Functions workflow - Complete. Successful execution captured in screenshots.
- 24-hour SES reminders - Partial. Reminder loop works; production wait is 86400 seconds; test delivery currently uses SNS.
- Progress API - Complete. Protected API and 100% progress response tested.
- Encrypted S3 document collection - Complete. All three required document types are supported.
- Lambda file validation - Complete. Type/size validation and document verification logic implemented.
- HR SNS notification - Complete. Notification is sent when all required documents are verified.
- New-hire portal - Complete. CloudFront/S3 frontend is operational.
- HR dashboard - Complete. Stage status and complete-stage actions are available.
- Execution screenshots - Complete. Captured separately.
- ER diagram - Complete. Included in this report.
- Cost estimate - Complete. Included below.
- Video demo - Pending. To be recorded separately.

14. Cost Estimate for 50 Onboardings per Month

- Assumptions: 50 new hires, 150 documents, approximately 1,000 API calls, 1,500 Lambda invocations, 1,500 Step Functions transitions, 50 Cognito monthly active users and low-volume frontend traffic.
- AWS Lambda: approximately 1,500 invocations - negligible cost at this scale.
- Amazon API Gateway HTTP API: approximately 1,000 calls - approximately $0.00-$0.01.
- AWS Step Functions Standard: approximately 1,500 transitions - approximately $0.00 at low usage.
- Amazon DynamoDB On-Demand: a few thousand reads and writes - typically below $0.05.
- Amazon S3: approximately 0.75 GB of new documents per month - approximately $0.02-$0.10, depending on requests and storage duration.
- Amazon Cognito: approximately 50 monthly active users - very low cost and may be covered by applicable free allowances.
- Amazon SES: approximately 250 messages - typically only a few cents, depending on region and usage.
- Amazon SNS: approximately 50 HR notifications - negligible at this scale.
- Amazon CloudFront: low portfolio/demo traffic - typically negligible or within applicable allowances.
- Amazon CloudWatch: small log volume - approximately $0.00-$0.50 depending on retention and ingestion.
- Estimated total: approximately $0-$2 per month under light usage; a $5 per month ceiling is a conservative budget estimate for the project.

15. Limitations and Future Improvements

- Complete SES-based login-link and 24-hour reminder delivery.
- Add Cognito groups and resource-level authorization for NEW_HIRE and HR_ADMIN roles.
- Allow the HR dashboard to list and select multiple employees instead of using one demonstration employee.
- Derive employee identity from authenticated claims instead of a hard-coded demonstration ID.
- Reconcile manually updated Lambda and Step Functions resources into the CloudFormation packaging and CI/CD process.
- Use a private S3 REST origin with CloudFront Origin Access Control for stronger frontend-bucket security.

16. References

1. *AWS Documentation - https://docs.aws.amazon.com/*
2. *AWS Lambda Documentation - https://docs.aws.amazon.com/lambda/*
3. *Amazon DynamoDB Documentation - https://docs.aws.amazon.com/dynamodb/*
4. *Amazon Cognito Documentation - https://docs.aws.amazon.com/cognito/*
5. *AWS Step Functions Documentation - https://docs.aws.amazon.com/step-functions/*
6. *Amazon S3 Documentation - https://docs.aws.amazon.com/s3/*

7. *Amazon API Gateway Documentation - https://docs.aws.amazon.com/apigateway/*
8. *Amazon Simple Notification Service Documentation - https://docs.aws.amazon.com/sns/*
9. *Amazon Simple Email Service Documentation - https://docs.aws.amazon.com/ses/*
10. *Amazon CloudFront Documentation - https://docs.aws.amazon.com/cloudfront/*
11. *AWS CloudFormation Documentation - https://docs.aws.amazon.com/cloudformation/*
12. *Amazon CloudWatch Documentation - https://docs.aws.amazon.com/cloudwatch/*